

What is the relation between magnification and focal length?

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8 Tech in Chemical Engineering, Indian Institute of Technology, Bombay (IITB) (Graduated 1975) · 5y

What is the relation between magnification and focal length?

For a lens

The lens formula is $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ and the magnification is, $M = \frac{v}{u}$.

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{u+f}{uf}$$
$$\Rightarrow v = \frac{uf}{u+f} \Rightarrow M = \frac{v}{u} = \frac{f}{u+f}$$

It may be noted that for a convex lens, u is always negative and f is always positive. For a concave lens, both u and f are always negative.

This is the relation between the magnification and focal length with the distance of the object ... (more)

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 **Chhavi Goenka** · Follow

Designer of optical instruments · Upvoted by RAJA, M.Sc Physics, Indian Institutes of Technology... · By

The relationship between magnification and focal length for different optical instruments or devices is different.

For a single thin lens, magnification, M, can be written as:

$$M = f/(f-d)$$

Here f is the focal length of the thin lens and d is the distance between the lens and object.

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 **Bill Otto** · Follow

Studied Physics & Chemistry at The University of Alabama in Huntsville (Graduated 1976) · 5y

Originally Answered: How do zoom, magnification, and focal length relate?

Let me assume you know what a focal length is so that I do not need to describe that.

Zoom is not well defined, but loosely, it is the ratio of magnifications from the shortest focal length to the longest.

For example, I have a lens that zooms from 18mm focal length to 55 mm focal length. This could be called a 3 to 1 zoom or a 3.05:1 zoom. I also have a 120mm to 600mm zoom lens. I pull it out for special shots like eclipses or wildlife. It is a 5:1 zoom.

You might find a point and shoot camera with a 30:1 zoom. I guarantee the image quality will not be that great, but it is very handy because it... (more)

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 **Assistant**

Bot · Aug 17

The relationship between magnification and focal length can be understood through the lens formula and the concepts of image formation. In optics, magnification (M) is defined as the ratio of the height of the image (h_i) to the height of the object (h_o):

$$M = \frac{h_i}{h_o}$$

For thin lenses, magnification is also related to the object distance (d_o) and the image distance (d_i):

$$M = -\frac{d_i}{d_o}$$

Focal Length and Magnification

1. **Thin Lens Formula:** The relationship between the object distance (d_o), image distance (d_i), and focal length (f) of a lens is given by the lens formula:

$$\frac{1}{f} =$$

... (more)

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 **Ludwig Keck** · Follow

Engineer on photo products for most of my career, cameras and more. · 8y

Magnification is the ratio of the size of the image to the size of the real object. Since cameras involve optics, there is some physics and math involved, but we will keep this simple. The magnification is also the ratio of the distances lens- to-image and lens-to-object. Let's not worry about where on a lens barrel you would measure those distances. For objects well in front of the lens the lens to image distance will be a little larger than the focal length.

Let's do an example. The subject is 2000mm in front of the lens. That's a bit more than 6 feet. With your lens set to 50mm on the zoom ... (more)

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 **Chris Patmore** · Follow

Photographer, Author and cook. Not an academic. (2002–present) · 6y

Originally Answered: What is the relation between focal length and magnification?

This may not be in the spirit of Quora, but this explains it pretty well and even includes equations and diagrams. [Focal length - Wikipedia](#)

Or to put it simply: the focal length is the distance the light has to travel from the front element of the lens to the film or sensor. The "magnification" or angle of view is dependent on the size of the film or sensor in relation to the focal length, which is why APS-C sensors are described as having a crop factor of say 1.6 in relation to a full-frame sensor, which is based on 35mm film. This is why camera manufacturers will often write 35mm equivalent o... (more)

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 **Ruben Valenzuela** · Follow

BA in astrophysics, UC Berkeley c/o 1992 · 8y

Where real images are involved, there is no direct relationship to magnification and focal length. Magnification of the image compared to the object can be calculated by the ratio of object and image distance, but this number can vary anywhere below 1 or above one. When the object and image are the same distance, the magnification is 1 and the image distance happens to be twice the focal length (2f), but this changes when you move the object closer or farther away from this point. Moving the object farther away than 2f would make the image distance decrease causing a decrease in magnification.... (more)

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 **Ganesh Subramaniam** · Follow

Amateur Radio Operator, Amateur Astronomer · Updated 2y

Originally Answered: Does focal length affect magnification?

Yes, it does. Magnification is inversely related to the focal length of a lens: the longer the focal length, the higher the magnifying power. When the focal length is long, the angle of view becomes narrow, increasing magnification. When the focal length is short, the angle of view is wide, and so magnification is lower.

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 **J. Pepper** · Follow

Engineer, Artist, Maker, Writer, Teacher, Veteran, Father · 4y

Originally Answered: How does the focal length affect magnification?

On a full-frame camera, 50 mm gives 1:1 magnification. Relative magnification of a lens is the focal length over 50 - e.g. a 100mm lens gives 2x magnification (100/50 = 2).

On a 1.6 crop camera, you get 1:1 magnification at 31 mm so on that camera a 62mm lens would give 2x magnification.

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 **Michel Ibester** · Follow

Works at Self-Employment · 2y

Originally Answered: Does focal length affect magnification?

yes it does the longer the focal length the more it magnifies, a telescope with a focal length of 1500mm fitted with an eyepiece of 10mm magnifies 150x (1500 divided by 10=150) the same eyepiece on a shorter telescope focal length 1000mm would be (1000 divided by 10=100)in other words focal length of telescope or indeed binoculars divided by the focal length of the eyepiece is the given magnification , and of course it is not just a question of just upping the eyepiece , because short focal lengths give poor resolution regardless to the make ,there is no substitute for length

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 **expert on this stuff, probably because Ive had so much of it happen to me** · 8y

The magnification is how zoomed in it is. The focal length, is how far away the part that it's focused on is.

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Originally Answered: Does focal length affect magnification?

Magnifying power is inversely related to the focal length of a lens: the bigger the focal length, the lower the magnifying power. The longer the focal length, the narrower the angle of view

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